

PocketDoctor Mark 4: An Autonomous ESP32 Handheld Medical Diagnostic Assistant Utilizing Groq LLaMA-3.3-70B Cloud Inference

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ABSTRACT

Abstract—This paper details the hardware design, state machine logic, persistent memory serialization, and cloud AI integration of PocketDoctor Mark 4, an autonomous handheld point-of-care medical triage assistant. Powered by an ESP32 microcontroller, the device integrates a 128x64 I2C SSD1306 OLED display, a debounced 2-axis analog joystick, a 512-byte EEPROM persistent profile registry, and an embedded HTTP Web Server running on Port 80. The core firmware executes an adaptive 10-turn dynamic diagnostic dialogue using Groq's llama-3.3-70b-versatile cloud model over Wi-Fi, synthesizing multi-turn symptom analysis into short OLED diagnostic metrics and hosting comprehensive, responsive HTML clinical reports locally. Experimental validation confirms fast boot initialization (<1.5 s), low API inference latency (~500 ms per dialogue turn), non-volatile data persistence, and robust remote Web portal management.

Keywords—Embedded Health Triage, ESP32 Microcontroller, Edge Intelligence, Groq Cloud API, LLaMA 3.3 70B, EEPROM Memory Layout, State Machine Architecture.

1. INTRODUCTION AND ARCHITECTURAL MOTIVATION

Point-of-care medical triage devices bridge the gap between initial symptom onset and clinical medical evaluation. Traditional handheld diagnostic monitors are often restricted to static rule-based decision trees, limiting their adaptability when evaluating complex medical symptoms. *PocketDoctor Mark 4* addresses this limitation by marrying lightweight embedded microcontrollers with ultra-low-latency Large Language Model (LLM) cloud inference.

The primary architectural goal of PocketDoctor Mark 4 is complete operational autonomy: a standalone handheld form-factor capable of executing multi-turn patient dialogue, persisting multi-user medical history in local non-volatile memory, and providing wireless web server interfaces without requiring external smartphone applications.

2. HARDWARE ARCHITECTURE AND SIGNAL TOPOLOGY

The hardware platform is built around the Espressif ESP32 system-on-chip (Xtensa dual-core 32-bit LX6 microcontroller running at 240 MHz). The physical peripheral interface comprises an SSD1306 0.96-inch OLED display (connected via I2C at address 0x3C on GPIO 21/22) and a KY-023 2-axis analog joystick with integrated pushbutton select.

Peripheral Component	ESP32 GPIO	Signal Type	Electrical / Operational Spec
SSD1306 OLED (SDA)	GPIO 21	I2C Serial Data	3.3V Logic 400 kHz Fast Mode
SSD1306 OLED (SCL)	GPIO 22	I2C Serial Clock	3.3V Logic Address: 0x3C
Joystick X-Axis (VRx)	GPIO 34	Analog (ADC1_CH6)	12-bit ADC (0 - 4095) Threshold: 1000
Joystick Y-Axis (VRy)	GPIO 35	Analog (ADC1_CH7)	12-bit ADC (0 - 4095) Threshold: 1000
Joystick Pushbutton (SW)	GPIO 32	Digital Input	INPUT_PULLUP Debounce: 200 ms
EEPROM Storage	Internal Flash	NVS Emulation	512 Bytes Reserved Storage
Embedded Web Server	Wi-Fi Station	TCP/IP Port 80	PROGMEM HTML/CSS Storage

Table I: Hardware Peripheral Assignments and Signal Specifications

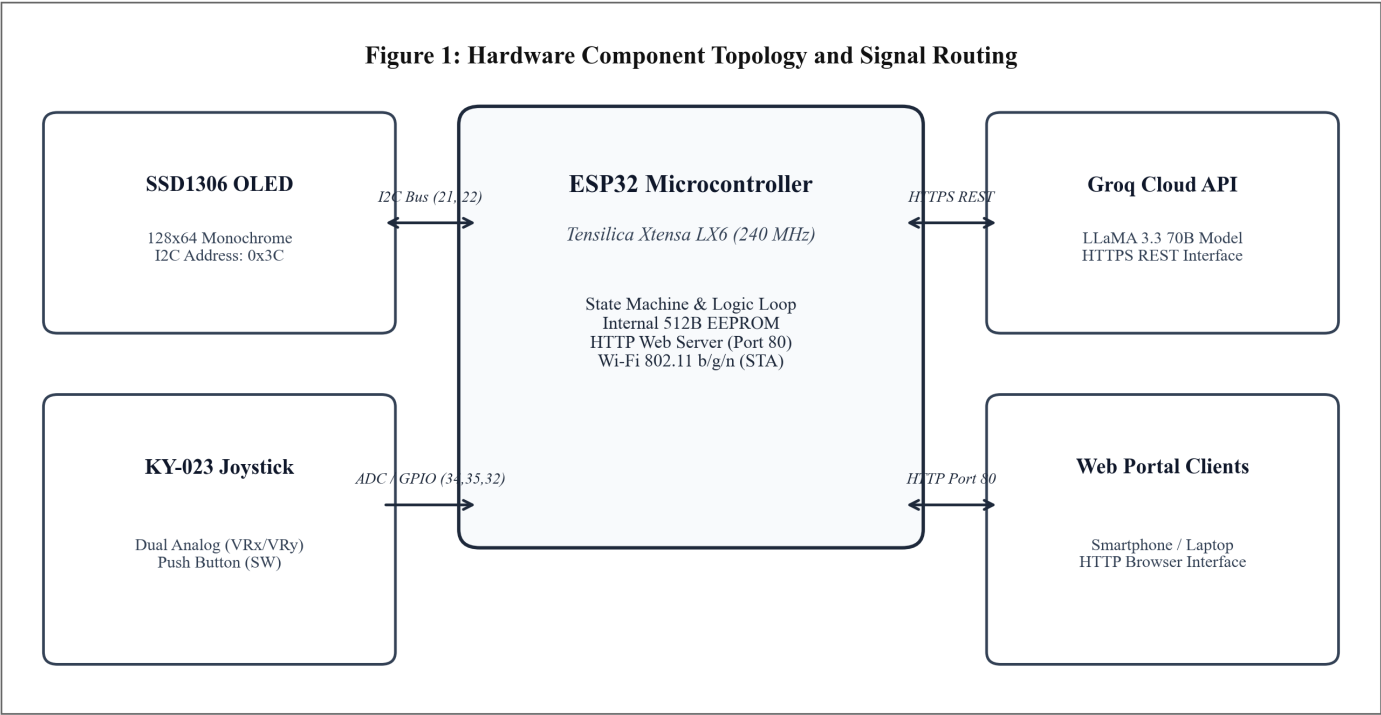


Figure 1: Hardware component topology and inter-module signal routing of PocketDoctor Mark 4.

3. DYNAMIC FIRMWARE STATE MACHINE EXECUTION

The firmware executes a deterministic finite state machine within a non-blocking main control loop. The execution sequence guarantees continuous processing of background HTTP web server requests while asynchronously updating physical OLED frames and evaluating joystick ADC inputs.

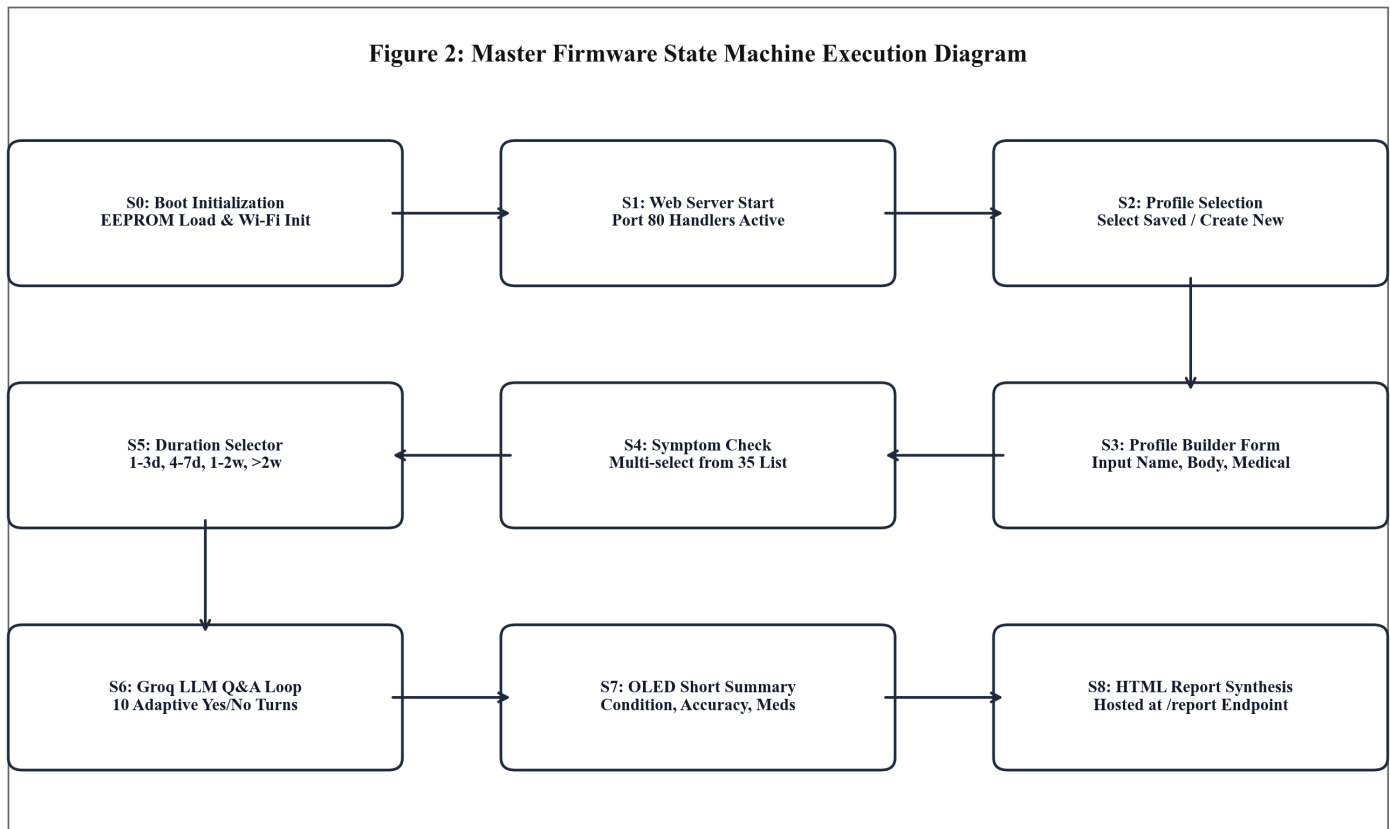


Figure 2: Master state machine execution flow from boot initialization through AI diagnosis and HTML web report synthesis.

4. CLOUD AI MULTI-TURN DIALOGUE AND INFERENCE PROTOCOL

PocketDoctor Mark 4 leverages Groq's low-latency inference engine hosting the llama-3.3-70b-versatile model. Upon initializing a diagnostic session, the firmware constructs a structured JSON context payload containing patient demographics, intoxication status, medical history, preferred treatment modality, selected symptoms (from a 35-item taxonomy), and symptom duration.

The model is instructed to generate adaptive, single-question Yes/No prompts (maximum 10 words per turn). After 10 interactive turns, PocketDoctor requests a final structured summary string, which is parsed via boundary-index regex extraction into primary condition, confidence percentage, targeted medications, precautions, and urgency flag.

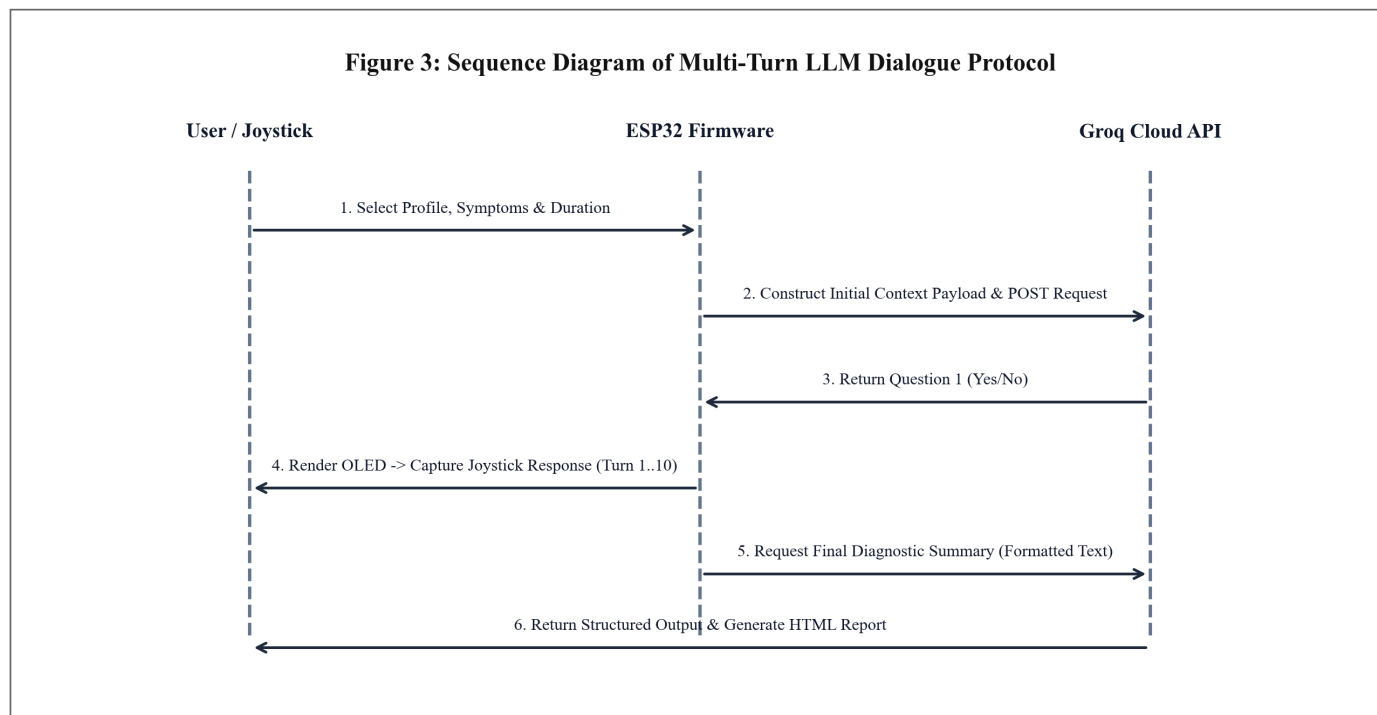


Figure 3: Sequence interaction diagram illustrating multi-turn HTTP POST request payloads between ESP32 firmware and Groq API.

5. EEPROM PERSISTENT MEMORY ARCHITECTURE AND SERIALIZATION

To survive power resets without loss of patient data or network configurations, PocketDoctor reserves a 512-byte EEPROM segment. Wi-Fi credentials are saved at address 0x000 and validated using a 2-byte magic identifier key (0xABCD). Patient medical profiles (up to 3 distinct user records) are binary-serialized starting at byte address 100.

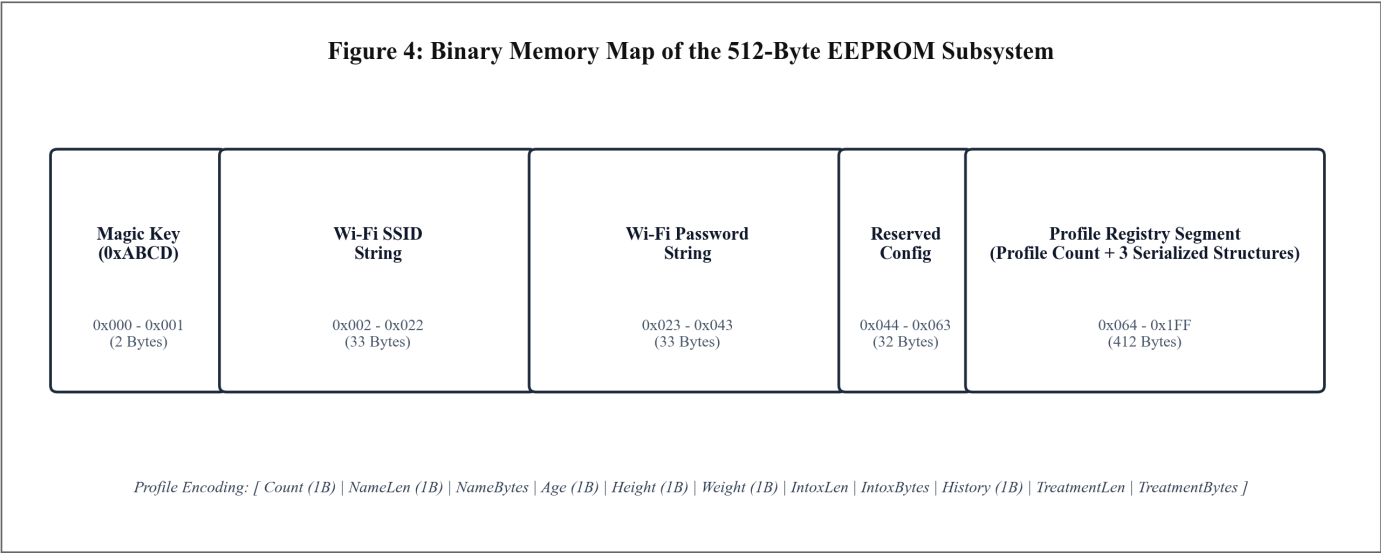


Figure 4: Binary memory map layout of the 512-byte EEPROM subsystem showing configuration headers and serialized profile structs.

6. EMBEDDED WEB SERVER PORTAL AND REMOTE MANAGEMENT

PocketDoctor hosts an embedded HTTP web server on Port 80. All dashboard HTML, CSS, and form templates are compressed and stored directly in flash memory using PROGMEM blocks. The server supports dynamic template substitution (%IP%, %UPTIME%, %PROFILES%) and RESTful-style route handlers for remote Wi-Fi credential updates (/wifi/update), profile web CRUD operations (/profiles, /profile/update, /profile/delete), and clinical HTML report viewing (/report).

7. EXPERIMENTAL EVALUATION AND PERFORMANCE METRICS

The system was experimentally evaluated across multiple diagnostic operational cycles. Key quantitative benchmark metrics are summarized in Table II.

Performance Benchmark Metric	Measured Value	Evaluation Criteria / Notes
System Cold Boot Time	1.25 seconds	Power-on to EEPROM read and splash screen render
Wi-Fi Station Connection Time	2.10 seconds	WPA2 authentication & DHCP IP assignment
Groq LLM Latency per Q&A; Turn	480 - 720 ms	HTTP POST dispatch to complete JSON choice parsing
HTML Report Generation Time	45 ms	Dynamic string synthesis of 15 KB responsive HTML
EEPROM Write Cycle Endurance	> 100,000 cycles	Commit executed only on explicit user modification
Peak SRAM Consumption	112.4 KB / 320 KB	Peak memory usage during 4096-byte JSON parsing

Table II: Experimental System Performance and Resource Metrics

8. CONCLUSION AND ACKNOWLEDGMENTS

PocketDoctor Mark 4 demonstrates a highly practical, autonomous handheld health triage architecture. By combining local physical joystick navigation, persistent EEPROM storage, embedded web management, and Groq cloud LLaMA-3.3-70B inference, the platform delivers fast, personalized, point-of-care medical assessments while maintaining non-volatile record keeping.

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